Solving AAS and ASA Triangles with the Law of Sines Answer Key

Precalculus

Quick Answers

1. 75	4. 16.0	7. 93.47	10. 436.23
2. 65	5. 14.25	8. 81.14	
3. 18.03	6. 9.77	9. 82.61	

Common wrong answers

3. *If they got 7.99: flipped the ratio and paired the known side with the angle it is not opposite*
5. *If they got 40.42: inverted the ratio, dividing by the sine of the angle opposite the unknown side*
8. *If they got 88.36: used the given angle at the other end of the baseline instead of the third angle in the denominator*

1. Given $A = 42°$, $B = 63°$, $a = 10$; find C.

Solution: No law of sines is needed — the angle sum settles it.

$$C = 180° - 42° - 63°$$

$$\boxed{C = 75°}$$

The side $a = 10$ is not used here; it is the pair that later problems build on.

2. Given $A = 35°$, $C = 80°$, $b = 12$; find B.

Solution: $B = 180° - 35° - 80°$, so

$$\boxed{B = 65°}$$

The given side b sits between the two given angles A and C, so this data is **ASA**. Once B is known, b and B form a complete pair and every other part of the triangle is reachable.

3. AAS: $A = 40°$, $B = 75°$, $a = 12$; find b.

Solution: Side a is opposite the given angle A, so the pair $a/\sin A$ is already complete — the law of sines applies with no extra step.

$$\frac{b}{\sin B} = \frac{a}{\sin A}$$
$$b = \frac{12\sin 75°}{\sin 40°} = \frac{12(0.965926)}{0.642788} = 18.0326\ldots$$

$$\boxed{b \approx 18.03}$$

Reasonable: $B > A$, so b must come out larger than $a = 12$.
Watch for: 7.99 — that is $12\sin 40°/\sin 75°$, the ratio built upside down.

4. ASA: $A = 52°$, $B = 48°$, $c = 20$; find a.
Solution: The given side c lies between the given angles, so no pair is complete. Find C first.

$$C = 180° - 52° - 48° = 80°$$

Now c and C are a pair:

$$\begin{aligned} \frac{a}{\sin A} &= \frac{c}{\sin C} \\ a &= \frac{20 \sin 52°}{\sin 80°} = \frac{20(0.788011)}{0.984808} = 16.0033\ldots \end{aligned}$$

$$\boxed{a \approx 16.00}$$

5. AAS: $B = 105°$, $C = 35°$, $b = 24$; find c.
Solution: Side b is opposite the given angle B, so the pair $b/\sin B$ is complete and c is one proportion away.

$$\begin{aligned} \frac{c}{\sin C} &= \frac{b}{\sin B} \\ c &= \frac{24 \sin 35°}{\sin 105°} = \frac{24(0.573576)}{0.965926} = 14.2514\ldots \end{aligned}$$

$$\boxed{c \approx 14.25}$$

Reasonable: C is the smallest angle, so c must be the shortest side.
Watch for: 40.42 — that is the reciprocal set-up, and it wrongly makes the side opposite the $35°$ angle the longest in the triangle.

6. ASA: $B = 38°$, $C = 71°$, $a = 15$; find b.
Solution: The given side a is opposite A, which is *not* given, so this is ASA and A must come first.

$$A = 180° - 38° - 71° = 71°$$

Set up the proportion before computing:

$$\begin{aligned} \frac{b}{\sin B} &= \frac{a}{\sin A} \\ b &= \frac{15 \sin 38°}{\sin 71°} = \frac{15(0.615661)}{0.945519} = 9.7670\ldots \end{aligned}$$

$$\boxed{b \approx 9.77}$$

Note $A = C = 71°$, so the triangle is isosceles with $a = c$.

7. Surveying, AAS: $A = 28°$, $B = 47°$, $a = 60$ m; find b.
Solution: The measured distance a is opposite the measured angle A, so the data is AAS and the pair is ready to use.

$$\begin{aligned} b &= \frac{a \sin B}{\sin A} = \frac{60 \sin 47°}{\sin 28°} \\ &= \frac{60(0.731354)}{0.469472} = 93.4694\ldots \end{aligned}$$

$$\boxed{b \approx 93.47 \text{ m}}$$

Units carry through unchanged: the ratio $\sin B/\sin A$ is dimensionless, so metres in gives metres out.

8. Baseline, ASA: $A = 63°$, $B = 41°$, $c = 120$ m; find b.
Solution: The baseline c joins the two stations, so it lies between the two measured angles — ASA. The angle opposite the requested side b is $B = 41°$, and the complete pair must be built from the third angle.

$$C = 180° - 63° - 41° = 76°$$

$$\begin{aligned} \frac{b}{\sin B} &= \frac{c}{\sin C} \\ b &= \frac{120 \sin 41°}{\sin 76°} = \frac{120(0.656059)}{0.970296} = 81.1372\ldots \end{aligned}$$

$$\boxed{b \approx 81.14 \text{ m}}$$

Watch for: 88.36 — that comes from dividing by $\sin 63°$, pairing the baseline with a station angle instead of with the angle at the transmitter.

9. Two steps, AAS: $A = 33°$, $B = 58°$, $a = 45$; find c.
Solution: The pair $a/\sin A$ is complete, but c needs the angle opposite it, which is not given — so the third angle is the required first step.

$$C = 180° - 33° - 58° = 89°$$

$$\begin{aligned} \frac{c}{\sin C} &= \frac{a}{\sin A} \\ c &= \frac{45 \sin 89°}{\sin 33°} = \frac{45(0.999848)}{0.544639} = 82.6109\ldots \end{aligned}$$

$$\boxed{c \approx 82.61}$$

C is nearly a right angle, so c is nearly the hypotenuse of an almost-right triangle; the check $45/\sin 33° \approx 82.62$ confirms the size.

10. Triangulation: $A = 57°$, $B = 49°$, $c = 500$ **m.**
(a) Find a**.** The shoreline c joins the two posts and lies between the two measured angles, so this is ASA. Find the angle at the buoy first.

$$C = 180° - 57° - 49° = 74°$$

$$\begin{aligned} \frac{a}{\sin A} &= \frac{c}{\sin C} \\ a &= \frac{500 \sin 57°}{\sin 74°} = \frac{500(0.838671)}{0.961262} = 436.2343\ldots \end{aligned}$$

$$\boxed{a \approx 436.23 \text{ m}}$$

(b) Model answer (open response — grade the reasoning). With AAS or ASA you know two angles, so the third is forced by the angle sum, and the shape of the triangle is completely fixed. The one given side then fixes the size: the law of sines produces exactly one positive value for each remaining side, because the ratio sin(known angle)/ sin(known angle) is a single number. So AAS and ASA data determine exactly one triangle. SSA is different: there you solve for an angle using sin, and the equation $\sin \theta = k$ with $0 < k < 1$ has *two* solutions between $0°$ and $180°$, namely θ and $180° - \theta$. Both can produce a valid triangle, so SSA may give two triangles, one, or none — that is the ambiguous case. The classmate is wrong: no extra measurement is needed here.
Look for: the angle sum forcing the third angle; the single given side setting the scale; and, for SSA, the two supplementary solutions of $\sin \theta = k$.